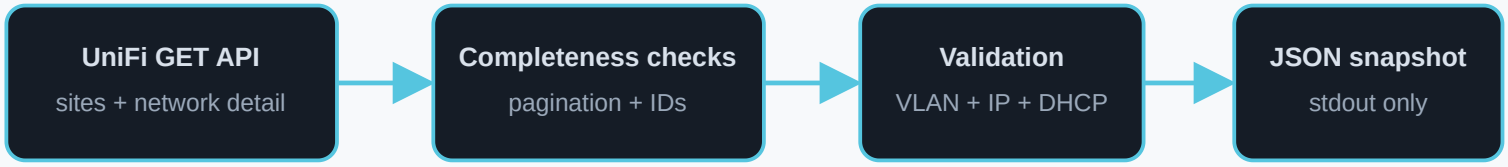
UniFi topology discovery

Revision 2 · 21 October 2026 · Status: read-only discovery and SoR comparison; no writer

This guide covers the UniFi topology collector in `deploy/unifi/unifi_topology.py`. It reads network pages and detail records through the UniFi Network Integration API, validates the returned topology, and prints a versioned JSON snapshot. An optional SoR comparison reads the current VLAN, network, subnet, DHCP range, and external-reference rows in a read-only transaction and reports correlation candidates and conflicts.

> **Current reconciliation gate.** The last recorded live UniFi read reported one Default network, VLAN 1, with `10.0.0.1/8` and DHCP pool `10.1.0.26–10.7.161.117`. This differs from the documented SolariNet segmented network model. The comparison command is read-only and reports the mismatch; it does not write to UniFi or the SoR. Do not apply a subnet or DHCP import until the source discrepancy is reconciled.

What the command does



- Follows paginated site and network lists until `totalCount` is satisfied. It checks response offsets, page counts, limits, stable UUIDs, and duplicate IDs.
- Fetches each network's detail record and validates enabled state, management mode, VLAN tag, IPv4 subnet and gateway, and DHCP server range.
- Rejects empty, malformed, partial, inconsistent, or ambiguous data. Multiple UniFi sites require an explicit site ID.
- Reports overlapping source subnets inside the snapshot. With `--compare-sor`, it also reads the active SoR VLANs, networks, subnets, DHCP ranges, and UniFi external references and reports exact matches, candidate matches, and conflicts.
- Uses GET requests for UniFi and a read-only SoR transaction. It has no SoR writer; names and VLAN tags are review hints, while a UniFi network external reference is the exact correlation.

Run a discovery

Run from the SolariNet repository root. The credential file is sourced into the process environment; the command never prints its contents. Do not enable shell tracing.

```
set -a
source /home/jason/Code/Solarinet/run/unifi.env
set +a
python3 deploy/unifi/unifi_topology.py --site-id 88f7af54-98f8-306a-a1c7-c9349722b1f6 --allow-insecure-tls
```

Set `UNIFI_SITE_ID` in the environment or pass `--site-id` when selecting a site. When `UNIFI_CA_FILE` points to a trusted controller CA certificate, omit `--allow-insecure-tls`.

```
python3 deploy/unifi/unifi_topology.py --site-id 88f7af54-98f8-306a-a1c7-c9349722b1f6 --allow-insecure-tls --compare-sor
```

To include the SoR comparison, configure `SOR_DB_HOST`, `SOR_DB_PORT`, `SOR_DB_NAME`, `SOR_DB_USER`, and `SOR_DB_PASSWORD` (or `SOR_DB_PASS`) through the approved protected environment, then add `--compare-sor`. The password is never placed on the command line or included in output. The comparison uses `SET TRANSACTION READ ONLY` and returns a `sor_comparison` object; without these variables, collection remains UniFi-only.

TLS note. The existing controller uses a certificate that is not trusted by this machine. The explicit `--allow-insecure-tls` option disables certificate verification for this GET session and prints a warning to standard error on every run. The UniFi API key is sent over that TLS connection and could be exposed to an active network interceptor. Prefer a trusted CA file and verified TLS; verified mode emits no insecure-TLS warning.

JSON is written to standard output. Redirect it to a protected local file if it needs to be retained. The snapshot contains internal site, network, subnet, gateway, and DHCP data.

Snapshot fields

FIELD	MEANING	VALIDATION
`schema_version`	Snapshot contract version; currently 1.	Required value 1.
`collected_at`	UTC completion time for the collection.	The UniFi API calls are sequential; this is not an atomic controller snapshot.
`sites[].id`, `networks[].id`	Native UniFi UUIDs used to correlate source records.	Canonical UUID format; duplicate IDs rejected.
`vlan_tag`	802.1Q VLAN tag from network configuration.	Integer 1–4094.

FIELD	MEANING	VALIDATION
`subnet`, `gateway_ip`	IPv4 prefix and configured gateway.	Valid IPv4 address and prefix; malformed or incomplete pairs fail.
`dhcp`	Mode and inclusive start/stop range when UniFi runs DHCP.	Both endpoints must be ordered, inside the subnet, outside network/broadcast, and must not contain the gateway. DHCP relay configurations fail closed until their server-address contract is represented.
`source_subnet_conflicts`	Overlaps among networks in this UniFi snapshot.	Any entry requires investigation before reconciliation.
`sor_comparison`	Optional read-only comparison with current SoR rows, including native network references, name/tag candidates, overlapping prefixes and overlapping DHCP ranges.	Rows with missing external references, ambiguous candidates, prefix/gateway/VLAN/DHCP differences, or source overlaps have `status="review_required"`. No actions are applied.

Current observed state

> **Read-only UniFi check · 21 October 2026.** The controller returned one site and one network. The normalized result reported VLAN 1, Default, subnet `10.0.0.0/8`, gateway `10.0.0.1`, DHCP mode `SERVER`, and pool `10.1.0.26–10.7.161.117`. The collector completed with no internal source-subnet overlaps.

A separate read-only query through the existing SoR sync account returned 16 active SoR subnets. Every row had `vlan_id=NULL` and `gateway_ip=NULL`.

SOR NETWORK	ACTIVE SUBNETS
core	`10.0.0.0/24`, `10.0.1.0/24`
production	`10.1.0.0/24`
IoT	`10.2.0.0/24`, `10.2.2.0/24`, `10.2.3.0/24`, `10.2.4.0/24`
peripherals	`10.3.0.0/24`
personal	`10.4.1.0/24`, `10.4.4.0/24`
workstations	`10.5.0.0/24`, `10.5.2.0/24`, `10.5.87.0/24`
netmgmt	`10.6.0.0/24`, `10.6.6.0/24`

SOR NETWORK	ACTIVE SUBNETS
security	`10.7.0.0/24`

The collector's pure prefix comparison reports 16 overlapping UniFi-to-SoR pairs, all `overlap` and none `same`. No SoR table rows were written. **Inference:** current SoR segmentation has no direct one-to-one CIDR match with UniFi's single Default /8. The live DHCP pool also spans several named SoR segments, so it cannot safely be split or reassigned without an authority decision.

The evidence confirms a source discrepancy but does not explain it. It could reflect an intentional broad UniFi network, incomplete SoR network metadata, or stale configuration/documentation. Do not enable automated reconciliation until the operator resolves which representation is authoritative.

Consistency note: UniFi site/network list and detail reads are separate API requests. `collected_at` marks when collection ended; it does not guarantee every detail came from one controller transaction. A future writer must include a freshness check and refuse a stale preview before apply.

Verification

```
python3 -m unittest -v deploy/unifi/test_unifi_topology.py deploy/unifi/test_unifipolld.py
python3 -m py_compile deploy/unifi/unifi_topology.py deploy/unifi/test_unifi_topology.py
```

The current fixture suite covers pagination, repeated and oversized pages, partial data, malformed site selection, UUID/detail matching, subnet and DHCP validation, overlap reporting, SoR read-only transaction behavior, reconciliation candidates, DHCP/gateway/VLAN mismatches, and the existing poller's behavior. CI discovers all `deploy/unifi/test_*.py` tests.

Remaining work

- Resolve the observed /8 versus segmented-prefix discrepancy and decide how the 16 SoR prefixes map to UniFi network identities.
- Determine whether SoR VLAN and gateway NULL fields are intentional incomplete planning data or missing imports.
- Use `--compare-sor` against an authorized read-only SoR connection and review the resulting correlation candidates. No automatic writer exists yet.
- After source authority is resolved, design the transactional, idempotent SoR writer with external references, audit provenance, deletion safeguards, and subnet locking before DHCP-range changes.
- Test writer replay, partial fetch failure, stale snapshots, allocator concurrency, and rollback against disposable MariaDB.
- Obtain cross-lab review before release. This guide documents discovery only; it does not claim complete UniFi-to-SoR integration.